

MTE 544 Assignment 1

Armaan Sengupta
20991907

1 Question 1: Mobile Manipulator

(a) The transformation matrix between two reference frames is defined as:

$$\mathcal{G}_b^a = \begin{bmatrix} R_b^a & q \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & x_b \\ \sin \theta & \cos \theta & y_b \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where $\bar{p}^a = \mathcal{G}_b^a \bar{p}^b$ with homogeneous coordinates $\bar{p} = \begin{bmatrix} p \\ 1 \end{bmatrix}$.

Starting from the end-effector frame $\{E\}$ and proceeding back to the global spatial frame $\{s\}$:

1. Transformation from $\{E\}$ to $\{2\}$ ($\mathcal{G}_E^2$): The end-effector frame $\{E\}$ has its origin displaced by link length l_2 along the X_2 axis. The rotation is fixed at $\psi = -\frac{\pi}{2}$:

$$\mathcal{G}_E^2 = \begin{bmatrix} \cos \psi & -\sin \psi & l_2 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\psi = -\frac{\pi}{2}} \mathcal{G}_E^2 = \begin{bmatrix} 0 & 1 & l_2 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

2. Transformation from $\{2\}$ to $\{1\}$ ($\mathcal{G}_2^1$): Reference frame $\{2\}$ is displaced by link length l_1 along the X_1 axis with joint rotation γ :

$$\mathcal{G}_2^1 = \begin{bmatrix} \cos \gamma & -\sin \gamma & l_1 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

3. Transformation from $\{1\}$ to $\{B\}$ ($\mathcal{G}_1^B$): Reference frame $\{1\}$ is displaced by distance l_B along the X_B axis with joint rotation α :

$$\mathcal{G}_1^B = \begin{bmatrix} \cos \alpha & -\sin \alpha & l_B \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

4. Transformation from $\{B\}$ to $\{s\}$ ($\mathcal{G}_B^s$): The base frame $\{B\}$ has its origin at (x, y) in the global reference frame with heading angle θ :

$$\mathcal{G}_B^s = \begin{bmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

5. Overall Transformation from $\{E\}$ to $\{s\}$ ($\mathcal{G}_E^s$): The overall transformation from frame $\{E\}$ to the global frame $\{s\}$ is:

$$\mathcal{G}_E^s = \mathcal{G}_B^s \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \quad (6)$$

Multiplying and simplifying yields:

$$\mathcal{G}_E^s = \begin{bmatrix} \sin(\theta + \alpha + \gamma) & \cos(\theta + \alpha + \gamma) & x + l_B \cos \theta + l_1 \cos(\theta + \alpha) + l_2 \cos(\theta + \alpha + \gamma) \\ -\cos(\theta + \alpha + \gamma) & \sin(\theta + \alpha + \gamma) & y + l_B \sin \theta + l_1 \sin(\theta + \alpha) + l_2 \sin(\theta + \alpha + \gamma) \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

(b) Given the numerical configuration:

$$\theta = -\frac{\pi}{2}, \quad \alpha = \frac{\pi}{6}, \quad \gamma = \frac{\pi}{4}, \quad l_1 = 1, \quad l_2 = 1, \quad l_B = 0.5, \quad x = 0.8, \quad y = 0.8$$

Substituting these values into the overall transformation matrix yields:

$$\mathcal{G}_E^s = \begin{bmatrix} -0.2588 & 0.9659 & 2.2659 \\ -0.9659 & -0.2588 & -0.8248 \\ 0 & 0 & 1.0000 \end{bmatrix} \quad (8)$$

To transform the given points $p_1^E = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $p_2^E = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ to the spatial frame $\{s\}$, we express them in homogeneous coordinates $\bar{p}_i^E = \begin{bmatrix} p_i^E \\ 1 \end{bmatrix}$ and apply $\bar{p}_i^s = \mathcal{G}_E^s \bar{p}_i^E$:

$$\bar{p}_1^s = \mathcal{G}_E^s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.2659 \\ -0.8248 \\ 1.0000 \end{bmatrix} \implies p_1^s = \begin{bmatrix} 2.2659 \\ -0.8248 \end{bmatrix} \quad (9)$$

$$\bar{p}_2^s = \mathcal{G}_E^s \begin{bmatrix} 0.5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.1024 \\ -1.5666 \\ 1.0000 \end{bmatrix} \implies p_2^s = \begin{bmatrix} 3.1024 \\ -1.5666 \end{bmatrix} \quad (10)$$

(c) A target object point is measured in the robot base frame $\{B\}$ as $q^B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, with homogeneous

coordinates $\bar{q}^B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

The kinematic transformation from the end-effector frame $\{E\}$ to the base frame $\{B\}$ is given by the chain of transformations as we have seen before in part (a):

$$\bar{q}^B = \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \bar{q}^E = \mathcal{G}_E^B \bar{q}^E \quad (11)$$

To express the target point in the end-effector frame $\{E\}$ for grasping, we invert this transformation. Note this means inverting the matrices as well as the order of multiplication:

$$\bar{q}^E = (\mathcal{G}_E^B)^{-1} \bar{q}^B = (\mathcal{G}_E^2)^{-1} (\mathcal{G}_2^1)^{-1} (\mathcal{G}_1^B)^{-1} \bar{q}^B \quad (12)$$

Evaluating the transformation with the numerical configuration from part **(b)**:

$$\bar{q}^E = (\mathcal{G}_E^2)^{-1}(\mathcal{G}_2^1)^{-1}(\mathcal{G}_1^B)^{-1} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.7418 \\ 0.3542 \\ 1.0000 \end{bmatrix} \quad (13)$$

Thus the location of the point in the end-effector frame $\{E\}$ is:

$$q^E = \begin{bmatrix} -0.7418 \\ 0.3542 \end{bmatrix} \quad (14)$$

2 Question 2: Differential-Drive Simulation

(a) i. Constant velocities

1.

2.

3.

ii. Velocity profiles

(b)

3 Question 3: 3-Omnivheel Simulation

(a)

(b) I. Rotation inputs

II. Designed control signals

1. Straight line with a slope of 60 degrees

2. Two-metre diameter circle